

Assessing and Managing Risk

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Learning objectives

Unit level learning objective(s)

4. discuss key health and safety responsibilities for engineers; and using recognised risk management tools create risk assessments to analyse a variety of project risks;

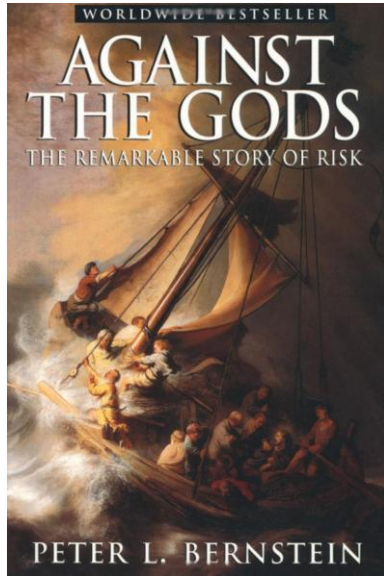
Session level learning objectives



1. **Create** a risk assessment using a recognised process whilst **identifying** frequency scores, consequence scores; and appropriate risk mitigations for operational, technical and project risks
2. **Generate** a scoring system for the classification of frequency and consequence
3. **Classify** risks using a criticality matrix



What is risk?



Quast, Pieter (c1605), *Soldiers Gambling with Dice*, Oil on Panel



Risk is... “the effect of uncertainty on objectives” [1]

[1] International Organization for Standardization (2018), *ISO 31000:2018 Risk Management – Guidelines*, ISO



Find BS and ISO standards (for free) at <http://www.bristol.ac.uk/library/find/standards/>

Types of objectives (and risks)



Safety



Scheduling



Financial



Performance

Quality



Environmental

Hazard... “something that has the potential to do harm”



Likelihood... “chance of something happening” [1]

Consequence... “the outcome of an event, affecting objectives” [1]

Mitigation... “a preventative or reactive action for reducing the likelihood of a risk, or consequence from risk”



Likelihood, Consequences, Mitigations,



Mitigation	Reduce Likelihood?	Reduce Consequence?
Climbing Helmet	✗?	✓
Rope	✓	✓



Methods in risk management



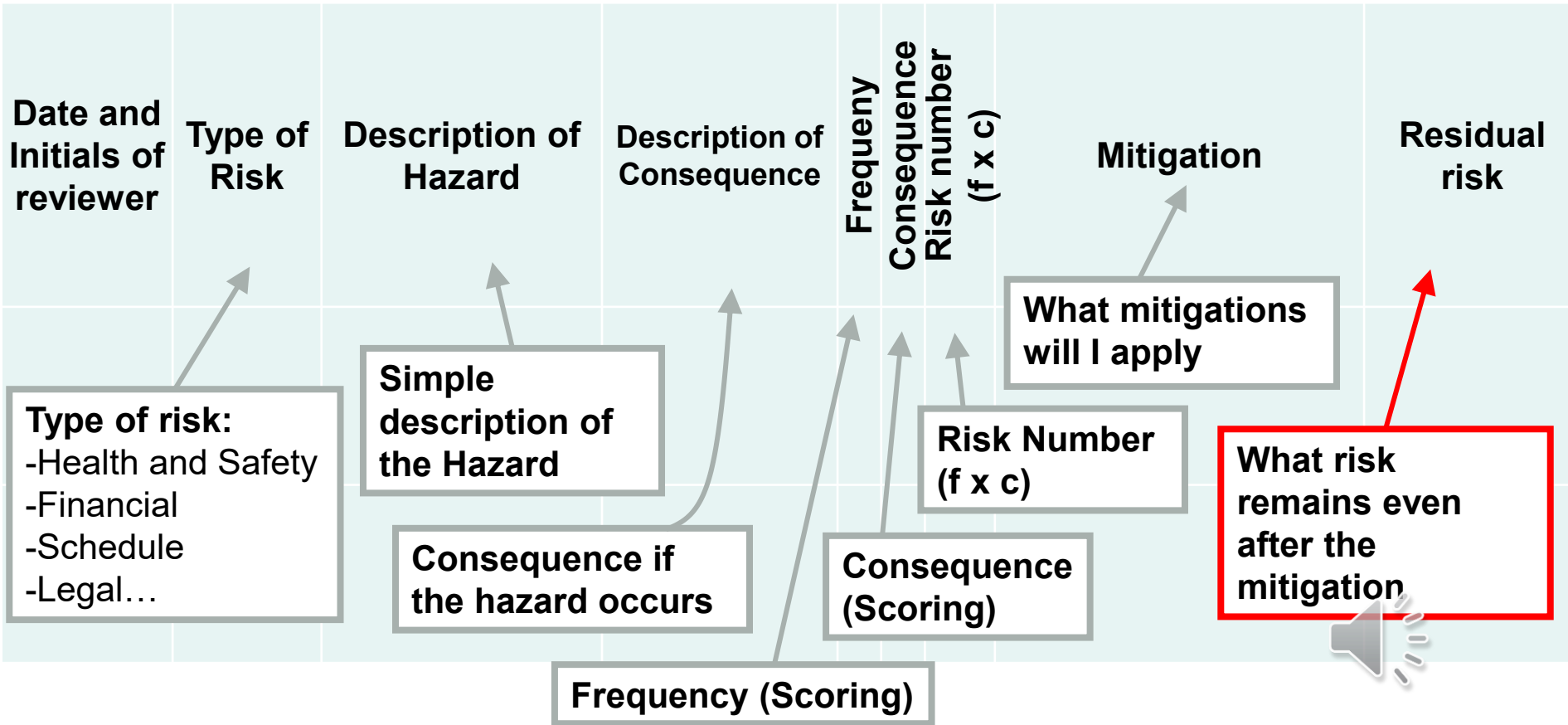
Completing a Risk Assessment

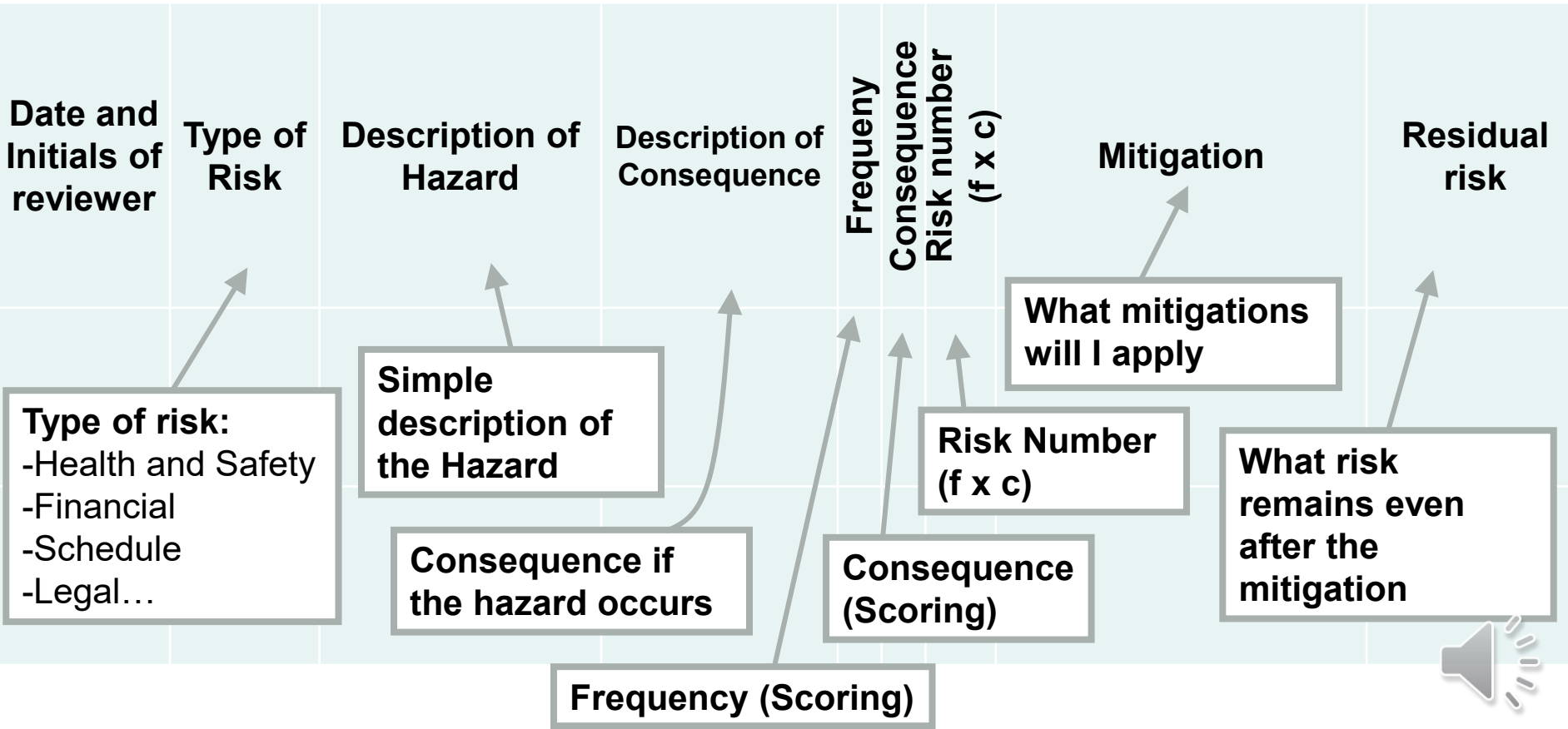
Date and Initials of reviewer	Type of Risk	Description of Hazard	Description of Consequence	Frequency	Consequence	Risk number (f x c)	Mitigation	Residual risk

An example Risk Assessment form

There are many others... Risk Register, DFMEA, PFMEA, IFMEA, FMECA







Assessing risk

Frequency category (score)	Classification term [1]
5	Almost Certain
4	Likely
3	Possible
2	Unlikely
1	Highly unlikely

Consequence category (score)	Classification term [1]
5	Catastrophic
4	Major
3	Moderate
2	Minor
1	Minimal

Catastrophic Safety event with the potential of **over 10 fatalities**, with **catastrophic long term environmental damage**

Catastrophic Safety event with the potential of **between two and 10 fatalities**, with major environmental impact resulting in Regulatory fines and current control measures are not suitable

Significant safety event with the potential of **between five major injuries and two fatalities**, with significant environmental impact that results in Regulatory intervention and it exceeds existing control measures

Significant Safety event with the potential of a **single major injury to five major injuries** with adverse environmental impact within a control period that can be mitigated using existing control measures

Minor Safety event with the potential to cause up to **20 minor injuries or a single major injury** and with environmental incidents that can be addressed using existing control measures

Assessing risk

			Consequence				
			1	2	3	4	5
			Minimal	Minor	Moderate	Major	Catastrophic
Frequency	5	Almost certain	5	10	15	20	25
	4	Likely	4	8	12	16	20
	3	Possible	3	6	9	12	15
	2	Unlikely	2	4	6	8	10
	1	Highly unlikely	1	2	3	4	5

Example Criticality Matrix

1-5 Negligible risk: Existing control measures are likely appropriate.

6-10 Tolerable risk: Additional control measures and mitigations required. Written action plan required.

11-25 Intolerable risk: Activity must halt. Resumption of activity requires approval from the safety board.

Treatment guidance (example only)



Summary



Thank you for your attention!

